

backend/core/max_power_runner.py Documentation (Roman Urdu)

Yeh file project list me sabse "extreme" tool hai jiska naam hai **Max-Power Persistent Runner**.

File Ka Maqsad (Purpose)

Aam tor par Ollama battery ya memory bachane k liye model ko RAM m wapis phenk deta hai aur GPU usage divide krleta ha. Magar Desktop users (Jin ke pas graphics card laga houy) is latency se chirr jatay hen. Yeh file Ollama ko force karti hai (`keep_alive=-1`) aur (`num_gpu=999`) daal kar k **SAARA** model permanent VRAM par dal do, aur har CPU thread max use kr do. Taa k AI generation bijli ki tezi se run ho.

Code Blocks aur Classes ki Tafseel

1. Hardcoded dictionaries (MAX_POWER_OPTIONS)

Kya karti hain?: Yahan commands ki override dictionary hai:

- `"num_gpu": 999` (GPU py full load krne ka hack)
- `"OLLAMA_KEEP_ALIVE": "-1"` (Aagosh m hmesha zinda rakho)
- `"CUDA_VISIBLE_DEVICES": "0"` (Sarf GPU #1 istamal karien)

2. class MaxPowerRunner

- **Kya karta hai?:** Yeh special executor (runner) ha!

Tafseel (Key functions):

- `warm_model(model)`: Pehlay kisi normal load hue model ko forcefully unload krata ha. Phr `MAX_POWER_OPTIONS` parameter k sth 0 token wali prompt bhejta hai. Js se model permanent GPU mein load/tayaar gari ki trha start khara rehta hai.
- `generate_stream()`: Model warm / load hou janay k baad jitni bhi chats stream hoti hein usmey MaxPower settings chipka deta ha proxy ki tarha.
- `_detect_cuda_library()`: Check karta ha `nvidia-smi` ko use kar kay k GPU mojud ha aur Ollama k konsay version k is py chalne ki probability zeada (e.g., `cuda_v12`) perfect banay gae.